Backward Bifurcation and Folds for the Quadratic $A\,I^2 + B(r)\,I + C(r) = 0$ under $\mathcal{R}_0(1-\epsilon) < 1 < \mathcal{R}_0$

Abstract

We study the equilibrium branch of the quadratic equation $AI^2+B(r)I+C(r)=0$, where B and C depend on the control parameter r through $\phi(r)=(\mu+\theta)(\mathcal{R}_0-r)/(r-\mathcal{R}_0(1-\epsilon))$, in the range $\phi\in[0,\infty)$ and under $\mathcal{R}_0(1-\epsilon)<1<\mathcal{R}_0$. We prove the following. (i) The numerator $N(r)$ of the discriminant B^2-4AC is a quadratic polynomial whose real roots, if any, all lie in $(\mathcal{R}_0\sigma,1]$ ($\sigma=1-\epsilon$); they exist if and only if $B(1)\le K(1-\mathcal{R}_0\sigma)$, i.e. $\beta\ge\beta^*$ for an explicit threshold β^*. (ii) If $B(1)<0$ (equivalently $\beta>\beta_0>\beta^*$) the bifurcation at $r=1$ is backward and *always* folds: the two positive branches $I_1(r)>I_2(r)$ exist exactly on $(r_2,1]$ and merge at the larger root $r_2\in(\mathcal{R}_0\sigma,1)$ of N; they can never stay real, positive and distinct down to $r=\mathcal{R}_0\sigma$. (iii) Regarded as a function $r=r(I)$ on $[I_a,I_b]$, where $r(I_a)=r(I_b)=\mathcal{R}_0$ and $I_a<0<I_b$ are explicit, the branch is strictly convex for $\beta\ge\beta^*$; its minimizer $I_+=\dfrac{-B(1)}{A}\Big/\Big(1+\sqrt{1-B(1)/(KD_1)}\Big)$ has the sign of $-B(1)$, so the bifurcation is backward if and only if the minimum occurs at a positive I. (iv) In the direct case the positive branch $I^+(r)$ is strictly increasing on $[1,\mathcal{R}_0]$ for every β; it is strictly concave if $\beta\ge\beta^*$, while for $\beta<\beta^*$ it is either concave or convex-then-concave with a single inflection point, the type being decided by the sign of an explicit quantity Θ.

1 Introduction

1.1 The problem

Let $\mu,\beta,\gamma,\theta,\psi>0$, $0<\epsilon<1$, $\sigma=1-\epsilon$, and

$$\begin{aligned}
A &= \beta\sigma(\mu+\gamma+\psi),\\
B &= (\mu+\psi)\big(-\beta\sigma+\theta+\mu\sigma+\mu+\sigma\phi(r)\big)+\gamma\big(\theta+\mu\sigma+\mu+\sigma\phi(r)+\sigma\psi\big),\\
C &= \frac{(\gamma+\mu)(\mu+\psi)(\theta+\mu+\phi(r))(1-r)}{\beta}, \qquad \phi(r)=(\mu+\theta)\frac{\mathcal{R}_0-r}{r-\mathcal{R}_0\sigma}.
\end{aligned}$$

The equilibria I for a given control value r are the roots of

$$A\,I^2 + B(r)\,I + C(r) = 0. \tag{1}$$

The range $\phi\in[0,\infty)$ corresponds to $r\in(\mathcal{R}_0\sigma,\mathcal{R}_0]$. Throughout we assume

$$\boxed{\mathcal{R}_0\sigma<1<\mathcal{R}_0} \qquad (\text{i.e. } D_1:=1-\mathcal{R}_0\sigma>0 \text{ and } \mathcal{R}_0>1). \tag{2}$$

At $r=1$ we have $C=0$ and the roots of (1) are $I=0$ and $I=-B(1)/A$, so $r=1$ is a transcritical point. The question is the nature of the bifurcation there (direct or backward) and the global shape of the endemic branch.

1.2 Summary of results

Write $K = (\gamma + \mu)(\mu + \psi)$. The formal statements are in the sections indicated.

R1 *Roots of the discriminant numerator* (Theorem 3.3). The numerator $N(r)$ of $B^2 - 4AC$ is a quadratic with positive leading coefficient. All its real roots lie in $(\mathcal{R}_0\sigma, 1]$ (never negative); it has real roots if and only if

$$B(1) \le K(1 - \mathcal{R}_0\sigma) \iff \beta \ge \beta^*,$$

and then it has two roots in $(\mathcal{R}_0\sigma, 1]$ (counted with multiplicity). There is never exactly one simple root in the interval. The condition is stated as an explicit threshold β^* in (13).

R2 *The bifurcation is always folded when backward* (Theorem 5.2). If $B(1) < 0$ (equivalently $\beta > \beta_0$, where $\beta_0 > \beta^*$), then with $r_1 < r_2$ the roots of N: $\mathcal{R}_0\sigma < r_1 < r_2 < 1$; there are two distinct positive equilibria $I_1(r) > I_2(r) > 0$ exactly for $r \in (r_2, 1)$; they merge at r_2 in a fold, and no positive equilibrium exists for $r < r_2$. In particular the case where the discriminant stays positive and the two branches reach $r = \mathcal{R}_0\sigma$ is impossible. The combination "$\beta < \beta^*$ and $B(1) < 0$" is impossible.

R3 *The branch as $r = r(I)$* (Section 6). Writing the equilibrium as $r = r(I)$ on $[I_a, I_b]$ with

$$I_{a,b} = \frac{-B_0 \mp \sqrt{B_0^2 + 4PK(\mathcal{R}_0 - 1)}}{2\beta\sigma(\mu + \gamma + \psi)} \qquad (B_0 = B|_{r=\mathcal{R}_0}),$$

the function is strictly convex for $\beta \ge \beta^*$ (Theorem 6.4). Its critical points are the roots of $A^2I^2 + 2AKD_1I + KD_1B(1) = 0$, and the minimizer is

$$I_+ = \frac{-B(1)}{A} \cdot \frac{1}{1 + \sqrt{1 - B(1)/(KD_1)}}, \qquad \operatorname{sign} I_+ = -\operatorname{sign} B(1).$$

Hence the bifurcation is backward if and only if the minimum of $r(I)$ occurs at a positive I (Theorem 6.5); when $\beta < \beta^*$, $r(I)$ has a pole inside $[I_a, I_b]$ and there is no critical point.

R4 *The positive branch in the direct case* (Section 7). For $r \in [1, \mathcal{R}_0]$ and $B(1) \ge 0$ the positive root $I^+(r)$ starts at 0, ends at I_b and is strictly increasing for every β. It is strictly concave if $\beta \ge \beta^*$. If $\beta < \beta^*$ it is concave when $\Theta \ge 0$, and when $\Theta < 0$ it is convex near $r = 1$ with at most one inflection point after which it is concave, where

$$\Theta = D_1B(1)^2 - pB(1) + pKD_1, \qquad p = \mathcal{R}_0\epsilon\sigma(\mu + \gamma + \psi)(\mu + \theta).$$

1.3 Numerical illustration

Take $\mu = 0.05$, $\gamma = \theta = \psi = 0.1$, $\mathcal{R}_0 = 3$, $\epsilon = 0.7$ (so $\sigma = 0.3$, $\mathcal{R}_0\sigma = 0.9$, $D_1 = 0.1$, $KD_1 = 0.00225$). The thresholds are $\beta^* \approx 5.933$ and $\beta_0 \approx 5.983$.

Backward bifurcation ($\beta = 8$). Then $B(1) \approx -0.0907 < 0$, the roots of N are $r_1 \approx 0.9655$, $r_2 \approx 0.9781$, and $I_1(1) = -B(1)/A \approx 0.1512$, $I_2(1) = 0$. The positive branches exist on $(r_2, 1]$ only (Figure **??**). In the $r(I)$ picture (Figure 1), $I_a \approx -0.00266$, $I_+ \approx 0.02036$, $I_b \approx 0.5289$, and $r(I_+) = r_2$ is the minimum.

Direct case. For $\beta = 3$, 1, 0.3 we have $B(1) = 0.134$, 0.224, $0.256 > KD_1$ (so $\beta < \beta^*$). The branch $I^+(r)$ is increasing in all three cases (Figure 2). For $\beta = 3$ and 1 it is convex up to $r \approx 1.143$ and 1.128 respectively and concave afterwards ($\Theta < 0$); for $\beta = 0.3$ it is concave throughout ($\Theta > 0$).

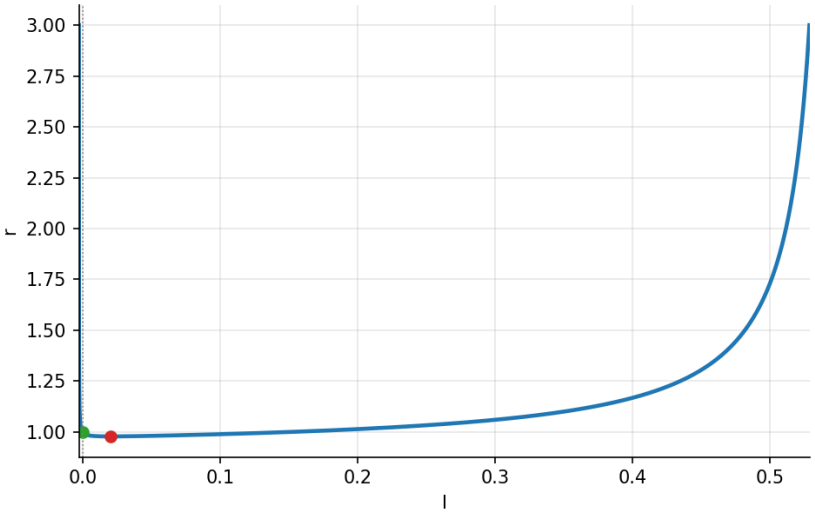


Figure 1: The same branch as $r = r(I)$ on $[I_a, I_b]$ for $\beta = 8$. The red dot is the minimum at $I_+ > 0$ (fold); the green dot is the transcritical point $(I, r) = (0, 1)$.

1.4 Organization

Section 2 fixes notation and computes the derivatives of ϕ, B, C. Section 3 derives the polynomial N and locates its roots (R1). Section 4 introduces β^*, β_0 and the three parameter regimes. Section 5 proves the fold theorem (R2). Section 6 studies $r = r(I)$ (R3). Section 7 treats the direct case (R4). Section 8 reports the numerical verification, and Section 9 collects remarks.

2 Preliminaries

2.1 Notation

$$D = r - \mathcal{R}_0\sigma, \quad D_1 = 1 - \mathcal{R}_0\sigma, \quad a = \mu + \gamma + \psi, \quad K = (\gamma + \mu)(\mu + \psi),$$
$$P = \sigma a(\mu + \theta), \quad Q = 4PK\epsilon\mathcal{R}_0, \quad M = \frac{K(\mu + \theta)\epsilon\mathcal{R}_0}{\beta}, \quad p = \mathcal{R}_0\epsilon P,$$
$$q = MD_1, \quad B_0' = a(\theta + \mu\sigma + \mu) + \gamma\sigma\psi, \quad B_0 = B_0' - \beta\sigma(\mu + \psi),$$

and $u = B_0 - P$, $v = P - \sigma B_0$.

2.2 Basic identities

B is affine in ϕ. The coefficient of ϕ in B is $(\mu + \psi)\sigma + \gamma\sigma = \sigma a$, so

$$B = \sigma a\,\phi + B_0. \tag{3}$$

Simplification of $\theta + \mu + \phi$ and C.

$$\theta + \mu + \phi = (\mu + \theta)\frac{D + \mathcal{R}_0 - r}{D} = \frac{\epsilon\mathcal{R}_0(\mu + \theta)}{D},$$

since $D + \mathcal{R}_0 - r = \mathcal{R}_0 - \mathcal{R}_0\sigma = \epsilon\mathcal{R}_0$. Hence

$$C = M\,\frac{1 - r}{D}, \qquad DC = M(1 - r), \qquad DB = P(\mathcal{R}_0 - r) + B_0 D. \tag{4}$$

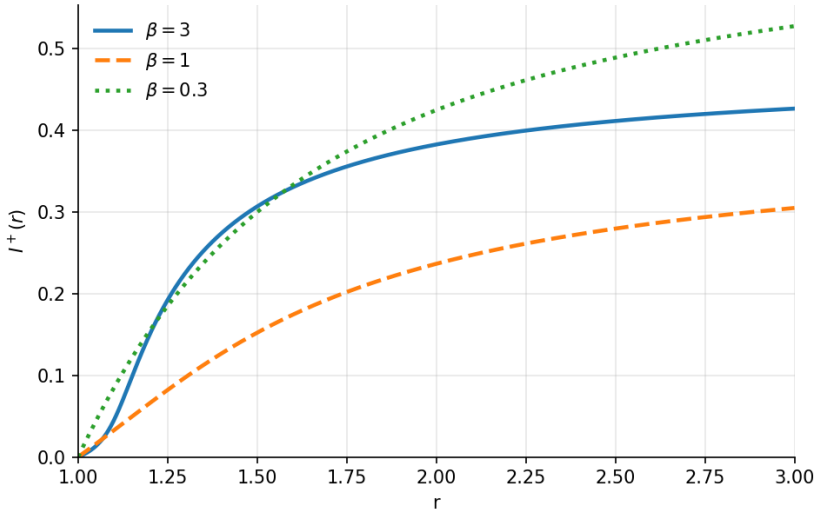


Figure 2: Positive branch $I^+(r)$ for $r \in [1, 3]$ in the direct case.

Values at $r = 1$. $\phi(1) = (\mu + \theta)(\mathcal{R}_0 - 1)/D_1$, so

$$L_1 := D_1 B(1) = P(\mathcal{R}_0 - 1) + B_0 D_1. \tag{5}$$

Algebraic relations. From $A = \beta\sigma a$ and $P = \sigma a(\mu + \theta)$, and from $\mathcal{R}_0\epsilon = (\mathcal{R}_0 - 1) + D_1$,

$$AM = pK, \qquad \sigma u + v = \epsilon P, \qquad u + \mathcal{R}_0 v = L_1, \qquad p + uD_1 = L_1. \tag{6}$$

2.3 Derivatives and signs

ϕ**.**

$$\phi'(r) = (\mu + \theta)\frac{-D - (\mathcal{R}_0 - r)}{D^2} = -\frac{\epsilon\mathcal{R}_0(\mu + \theta)}{D^2} < 0,$$

so ϕ decreases strictly from $+\infty$ to 0 on $(\mathcal{R}_0\sigma, \mathcal{R}_0]$.

B**.** By (3),

$$B'(r) = \sigma a\, \phi'(r) = -\frac{\epsilon\sigma\mathcal{R}_0(\mu + \theta)(\mu + \gamma + \psi)}{D^2} = -\frac{p}{D^2} < 0, \tag{7}$$

and $B \to +\infty$ as $r \to (\mathcal{R}_0\sigma)^+$.

C**.** From (4), $\frac{d}{dr}\frac{1-r}{D} = \frac{-D-(1-r)}{D^2} = -\frac{D_1}{D^2}$, so

$$C'(r) = -\frac{MD_1}{D^2} < 0, \qquad C''(r) = \frac{2MD_1}{D^3} > 0 \quad (D > 0). \tag{8}$$

Equivalently $C' = \frac{K}{\beta}[\phi'(1-r) - (\theta+\mu+\phi)]$, and the bracket equals $-\epsilon\mathcal{R}_0(\mu+\theta)[(1-r)+D]/D^2 = -\epsilon\mathcal{R}_0(\mu + \theta)D_1/D^2$ because $(1 - r) + D = D_1$ does not depend on r.

Sign of C. By (4), C has the sign of $(1 - r)/D$:

$$C > 0 \text{ on } (\mathcal{R}_0\sigma, 1), \qquad C(1) = 0, \qquad C < 0 \text{ for } r > 1 \text{ and for } r < \mathcal{R}_0\sigma. \tag{9}$$

(For $r < \mathcal{R}_0\sigma < 1$, $1 - r > 0$ and $D < 0$; for $r > 1$, $1 - r < 0$ and $D > 0$.)

3 The numerator of the discriminant

3.1 Derivation of $N(r)$

For $r \neq \mathcal{R}_0\sigma$ multiply $B^2 - 4AC$ by D^2. By (4), $(BD)^2 = [P(\mathcal{R}_0 - r) + B_0 D]^2$ and $4AC\,D^2 = 4\beta\sigma a\,M(1-r)D = Q(1-r)D$, because $4AM = 4pK$ and $p = \mathcal{R}_0\epsilon P$ give $4\beta\sigma aM = Q$. Hence

$$N(r) := D^2(B^2 - 4AC) = \big[P(\mathcal{R}_0 - r) + B_0 D\big]^2 - Q(1-r)D. \tag{10}$$

The bracketed linear part equals $ur + \mathcal{R}_0 v$. Expanding $Q(1-r)D = Q[-r^2 + (1+\mathcal{R}_0\sigma)r - \mathcal{R}_0\sigma]$,

$$\begin{aligned} N(r) &= c_2 r^2 + c_1 r + c_0, \\ c_2 &= u^2 + Q > 0, \quad c_1 = 2\mathcal{R}_0 uv - Q(1+\mathcal{R}_0\sigma), \quad c_0 = \mathcal{R}_0^2 v^2 + Q\mathcal{R}_0\sigma > 0. \end{aligned} \tag{11}$$

In particular the product of the roots of N is $c_0/c_2 > 0$: the two roots can never have opposite signs. Special values:

$$N(\mathcal{R}_0\sigma) = P^2\mathcal{R}_0^2\epsilon^2 > 0, \qquad N(1) = L_1^2 = D_1^2 B(1)^2 \geq 0, \tag{12}$$

the second because $C(1) = 0$.

3.2 Discriminant of N

Proposition 3.1.

$$c_1^2 - 4c_2c_0 = 16\,P^2K\,\epsilon^2\mathcal{R}_0^2\,D_1\big[KD_1 - B(1)\big].$$

Proof. From (11),

$$\begin{aligned} c_1^2 - 4c_2c_0 &= 4\mathcal{R}_0^2u^2v^2 - 4\mathcal{R}_0uvQ(1+\mathcal{R}_0\sigma) + Q^2(1+\mathcal{R}_0\sigma)^2 - 4(u^2+Q)(\mathcal{R}_0^2v^2 + Q\mathcal{R}_0\sigma) \\ &= Q^2D_1^2 - 4Q\mathcal{R}_0\Big[\sigma u^2 + (1+\mathcal{R}_0\sigma)uv + \mathcal{R}_0v^2\Big] = Q^2D_1^2 - 4Q\mathcal{R}_0(\sigma u + v)(u + \mathcal{R}_0 v). \end{aligned}$$

By (6), $\sigma u + v = \epsilon P$ and $u + \mathcal{R}_0 v = L_1 = D_1 B(1)$. With $Q = 4PK\epsilon\mathcal{R}_0$,

$$c_1^2 - 4c_2c_0 = 16P^2K^2\epsilon^2\mathcal{R}_0^2D_1^2 - 16P^2K\epsilon^2\mathcal{R}_0^2D_1B(1). \qquad \square$$

Since $D_1 > 0$, N has two real roots (with multiplicity) iff $B(1) \leq KD_1$, and two distinct real roots iff $B(1) < KD_1$.

3.3 Location of the roots

Lemma 3.2. *Under* (2), $N(r) > 0$ *for all* $r < \mathcal{R}_0\sigma$ *and all* $r > 1$, *and* $N(\mathcal{R}_0\sigma) > 0$.

Proof. For $r \neq \mathcal{R}_0\sigma$, $N = D^2(B^2 - 4AC)$. By (9), $C < 0$ for $r < \mathcal{R}_0\sigma$ and for $r > 1$, so $-4AC > 0$ and $B^2 - 4AC > 0$. The value at $\mathcal{R}_0\sigma$ is (12). $\square$

Theorem 3.3 (Roots of N)**.** *Under* (2), *every real root of* N *lies in* $(\mathcal{R}_0\sigma, 1]$*; in particular no root is negative.* N *has real roots if and only if* $B(1) \leq KD_1$. *If* $B(1) < KD_1$ *it has exactly two distinct real roots* $r_1 < r_2$ *in* $(\mathcal{R}_0\sigma, 1]$, *with* $r_2 < 1$ *if* $B(1) \neq 0$. *If* $B(1) = KD_1$ *it has one double root in* $(\mathcal{R}_0\sigma, 1]$. *A single simple root in* $(\mathcal{R}_0\sigma, 1)$ *never occurs.*

Proof. Lemma 3.2 excludes roots outside $(\mathcal{R}_0\sigma, 1]$. Proposition 3.1 gives the criterion for real roots. For $B(1) \neq 0$, $N(1) = L_1^2 > 0$ so $r_2 < 1$. If $B(1) = 0$ then $r = 1$ is a root and the other root, being in $(\mathcal{R}_0\sigma, 1]$ too, is < 1 (it is c_0/c_2 since the roots multiply to c_0/c_2, and $N > 0$ on $(1, \infty)$). The count is even because N is a convex quadratic positive at both ends of $[\mathcal{R}_0\sigma, 1]$ up to the endpoint case just discussed. $\square$

4 Thresholds and the three regimes

$B(1) = \sigma a\phi(1) + B_0$ is affine and strictly decreasing in β, since $B_0 = B_0' - \beta\sigma(\mu+\psi)$. Define

$$\beta^* = \frac{P(\mathcal{R}_0 - 1) + (B_0' - KD_1)D_1}{\sigma(\mu+\psi)D_1} \qquad \text{(where } B(1) = KD_1\text{)}, \tag{13}$$

$$\beta_0 = \frac{P(\mathcal{R}_0 - 1)/D_1 + B_0'}{\sigma(\mu+\psi)} \qquad \text{(where } B(1) = 0\text{)}, \tag{14}$$

so that

$$\beta_0 = \beta^* + \frac{KD_1}{\sigma(\mu+\psi)} > \beta^*.$$

Then $B(1) \le KD_1 \iff \beta \ge \beta^*$ and $B(1) \le 0 \iff \beta \ge \beta_0$. (If $\beta^* \le 0$ the first condition holds for every $\beta > 0$, and similarly for β_0.) The three regimes are:

- $\beta < \beta^*$**.** $N > 0$ everywhere, and $B(1) > KD_1 > 0$, so $B > 0$ on $(\mathcal{R}_0\sigma, 1]$ (as B is decreasing): both roots in I are negative for $r < 1$. The bifurcation is direct and there is no fold.
- $\beta^* \le \beta \le \beta_0$ ($0 \le B(1) \le KD_1$). N has real roots in $(\mathcal{R}_0\sigma, 1]$, but $B \ge 0$ on $(\mathcal{R}_0\sigma, 1)$, so roots in I are not positive there. The bifurcation is direct; a "hidden" fold sits at $I \le 0$ (Section 6).
- $\beta > \beta_0$ ($B(1) < 0$). The bifurcation is backward with a visible fold at $r_2 \in (\mathcal{R}_0\sigma, 1)$ (Section 5).

In particular the combination "$\beta < \beta^*$ and $B(1) < 0$" is impossible, since $B(1) < 0 < KD_1$ forces $\beta > \beta_0 > \beta^*$. The threshold β^* governs only whether N has real roots; the backward bifurcation itself is governed by β_0, because positive roots in I require $B < 0$.

5 The fold theorem

The roots of (1) are $I_{1,2} = \dfrac{-B \pm \sqrt{B^2 - 4AC}}{2A}$, with sum $-B/A$ and product C/A.

Lemma 5.1. *Let $r \in (\mathcal{R}_0\sigma, 1)$, so $A > 0$ and $C > 0$. The quadratic has two positive roots (counted with multiplicity) iff $B(r) < 0$ and $B^2 - 4AC \ge 0$. If $B(r) \ge 0$ both roots are negative or complex. At $r = 1$ the roots are 0 and $-B(1)/A$.*

Proof. The product $C/A > 0$ forces equal signs (or complex conjugates); the common sign is that of the sum $-B/A$. At $r = 1$, $C = 0$ and the equation factors as $I(AI + B(1)) = 0$. □

Theorem 5.2 (Fold)**.** *Assume* (2) *and $B(1) < 0$ (i.e. $\beta > \beta_0$). Let $r_1 < r_2$ be the roots of N. Then $\mathcal{R}_0\sigma < r_1 < r_2 < 1$, and:*

(a) B has exactly one zero $r_B \in (\mathcal{R}_0\sigma, 1)$, and $r_1 < r_B < r_2$.

(b) For $r \in (r_2, 1)$ the quadratic has two distinct positive roots $I_1(r) > I_2(r) > 0$, continuous in r. At $r = 1$: $I_2(1) = 0$, $I_1(1) = -B(1)/A > 0$.

(c) At $r = r_2$ the roots coincide: $I_1(r_2) = I_2(r_2) = -B(r_2)/(2A) > 0$. This is a fold (saddle-node), with $I_1 - I_2 \sim c\sqrt{r - r_2}$, $c > 0$, as $r \to r_2^+$.

(d) For $r \in (\mathcal{R}_0\sigma, r_2)$ there is no positive root: on (r_1, r_2) the roots are complex, and on $(\mathcal{R}_0\sigma, r_1]$ (where $B > 0$) they are real and negative.

Consequently the branches always merge in a fold at the larger root $r_2 \in (\mathcal{R}_0\sigma, 1)$ of N; they can never remain real, positive and distinct down to $r = \mathcal{R}_0\sigma$.

Proof. Since $B(1) < 0 < KD_1$, Proposition 3.1 gives two distinct real roots of N, and Theorem 3.3 places them in $(\mathcal{R}_0\sigma, 1)$ ($r_2 < 1$ as $B(1) \neq 0$).

(a) B is strictly decreasing, tends to $+\infty$ at $\mathcal{R}_0\sigma^+$ and $B(1) < 0$, so it has exactly one zero $r_B \in (\mathcal{R}_0\sigma, 1)$. At r_B, $B^2 - 4AC = -4AC < 0$ by (9), so $N(r_B) < 0$. As $N > 0$ outside $[r_1, r_2]$, $r_B \in (r_1, r_2)$.

(b) For $r \in (r_2, 1)$: $r > r_B$ gives $B < 0$, and $N > 0$ gives $B^2 - 4AC > 0$. Lemma 5.1 gives two distinct positive roots, continuous because $B^2 - 4AC > 0$. The values at $r = 1$ follow from the lemma.

(c) $N(r_2) = 0$ gives $B^2 = 4AC$ at r_2, and $B(r_2) < 0$ since $r_2 > r_B$, so both roots equal $-B(r_2)/(2A) > 0$. The roots of N are simple (positive discriminant), so $B^2 - 4AC = N/D^2$ has a simple zero at r_2 with positive slope, and $I_1 - I_2 = \sqrt{B^2 - 4AC}/A \sim c\sqrt{r - r_2}$.

(d) On (r_1, r_2), $N < 0$ gives complex roots. On $(\mathcal{R}_0\sigma, r_1]$, $r \leq r_1 < r_B$ gives $B > 0$, so by Lemma 5.1 the roots are not positive; the discriminant is ≥ 0 there, so they are real and negative. □

Remark 5.3. The alternative that the discriminant of N stays positive and two distinct positive branches reach $\mathcal{R}_0\sigma$ is excluded for two reasons: positive roots need $B < 0$ whereas $B \to +\infty$ at $\mathcal{R}_0\sigma^+$; and at the zero r_B of B the roots satisfy $I^2 = -C/A < 0$.

6 The branch as a function $r = r(I)$

6.1 Explicit form

Multiply (1) by D and use (4):

$$AI^2D + I\big[P(\mathcal{R}_0 - r) + B_0D\big] + M(1 - r) = 0,$$

which is *linear* in r. Collecting terms,

$$r\left(AI^2 + uI - M\right) = \mathcal{R}_0\sigma AI^2 - \mathcal{R}_0vI - M.$$

Using $\sigma u + v = \epsilon P$, polynomial division gives

$$r(I) = \mathcal{R}_0\sigma + \frac{p\,I + q}{\Delta(I)}, \qquad \Delta(I) := M - uI - AI^2. \tag{15}$$

In particular $r(0) = \mathcal{R}_0\sigma + q/M = 1$. The function has poles at the two zeros of Δ (one negative, one positive, since $\Delta(0) = M > 0$ and $A > 0$), where $r \to \pm\infty$, outside the physical range $\mathcal{R}_0\sigma < r \leq \mathcal{R}_0$. Because $r - \mathcal{R}_0\sigma = (pI + q)/\Delta$, with $p, q > 0$, the physical range corresponds to $pI + q > 0$ and $\Delta > 0$ on a pole-free piece.

6.2 The endpoints $I_a < 0 < I_b$

$r = \mathcal{R}_0$ means $\phi = 0$, where $B = B_0$ and $C = -K(\mu + \theta)(\mathcal{R}_0 - 1)/\beta$. Hence $r(I) = \mathcal{R}_0$ exactly at the roots of $AI^2 + B_0I - m = 0$, $m := K(\mu + \theta)(\mathcal{R}_0 - 1)/\beta = PK(\mathcal{R}_0 - 1)/A$:

$$I_{a,b} = \frac{-B_0 \mp \sqrt{B_0^2 + 4PK(\mathcal{R}_0 - 1)}}{2\beta\sigma(\mu + \gamma + \psi)}, \tag{16}$$

with $B_0 = (\mu + \gamma + \psi)(\theta + \mu\sigma + \mu) + \gamma\sigma\psi - \beta\sigma(\mu + \psi)$. The product of the roots is $-m/A < 0$, so $I_a < 0 < I_b$.

6.3 Absence of poles on $[I_a, I_b]$

Lemma 6.1. *Let $I_c := -KD_1/A < 0$.*

(i) $\Delta(I_b) > 0$ *and* $\Delta(I_a) = P(I_a - I_c)$.

(ii) If $B(1) < KD_1$ (i.e. $\beta > \beta^*$), then $I_c < I_a$, $\Delta > 0$ and $pI + q > 0$ on all of $[I_a, I_b]$, so $r(I)$ is smooth there and $\mathcal{R}_0\sigma < r(I) \le \mathcal{R}_0$.

(iii) If $B(1) > KD_1$ (i.e. $\beta < \beta^*$), then $I_a < I_c < 0$, so Δ vanishes somewhere in $(I_a, 0)$: $[I_a, I_b]$ contains a pole.

Proof. At $I = I_a, I_b$ we have $AI^2 + B_0 I = m$, so

$$\Delta(I) = M - (B_0 - P)I - AI^2 = M - m + PI = P\Big(I + \frac{KD_1}{A}\Big),$$

because $M - m = K(\mu + \theta)(\epsilon\mathcal{R}_0 - \mathcal{R}_0 + 1)/\beta = K(\mu + \theta)D_1/\beta = PKD_1/A$. This gives (i).

Let $G(I) = AI^2 + B_0 I - m$, a convex quadratic with roots $I_a < 0 < I_b$. Using $Am = PK(\mathcal{R}_0 - 1)$ and (5),

$$G(I_c) = \frac{K}{A}\Big[KD_1^2 - B_0 D_1 - P(\mathcal{R}_0 - 1)\Big] = \frac{KD_1}{A}\big(KD_1 - B(1)\big).$$

As $I_c < 0$: $I_c < I_a$ iff $G(I_c) > 0$ iff $B(1) < KD_1$; and $I_a < I_c < 0$ iff $B(1) > KD_1$. In case (ii), $\Delta(I_a) > 0$, and a concave quadratic positive at both endpoints is positive in between. Finally $r(I_a) = \mathcal{R}_0 > \mathcal{R}_0\sigma$ with $\Delta(I_a) > 0$ gives $pI_a + q > 0$, and $pI + q$ is increasing. □

6.4 Critical points of $r(I)$

Proposition 6.2. *Assume $B(1) \le KD_1$. The critical points of $r(I)$ are the roots of*

$$A^2 I^2 + 2AKD_1\, I + KD_1 B(1) = 0, \tag{17}$$

namely

$$I_\pm = \frac{-KD_1 \pm \sqrt{KD_1\big(KD_1 - B(1)\big)}}{A}, \tag{18}$$

and

$$\boxed{I_+ = \frac{-B(1)}{A} \cdot \frac{1}{1 + \sqrt{1 - \dfrac{B(1)}{KD_1}}}} \tag{19}$$

Hence $\operatorname{sign} I_+ = -\operatorname{sign} B(1)$, *with $I_+ = 0$ iff $B(1) = 0$, and for $B(1) < 0$, $0 < I_+ < -B(1)/(2A)$. If $B(1) < KD_1$ then $I_- < I_c < I_a$, so $I_- \notin [I_a, I_b]$.*

Proof. With $n = pI + q$, $dr/dI = 0$ iff $p\Delta - n\Delta' = 0$, i.e. $pAI^2 + 2qAI + pM + qu = 0$. Multiply by A/p and use (6): $qA/p = KD_1$, $AM = pK$, and $K(p + uD_1) = KL_1 = KD_1 B(1)$. This gives (17), whose roots are (18). Writing $\sqrt{x} - 1 = (x - 1)/(\sqrt{x} + 1)$ with $x = 1 - B(1)/(KD_1)$ gives (19). Finally $I_- = I_c - \sqrt{KD_1(KD_1 - B(1))}/A < I_c < I_a$ by Lemma 6.1(ii). □

Remark 6.3 (Critical points are folds). Let $F(r, I) = AI^2 + B(r)I + C(r)$ and $\Phi = DF$, which is linear in r with $\Phi_r = -\Delta$. On $\Phi = 0$, $dr/dI = \Phi_I/\Delta$, and $\Phi_I = DF_I$ (as $F = 0$). Hence, where $\Delta \ne 0$, $dr/dI = 0$ iff $F_I = 2AI + B = 0$, i.e. iff the quadratic has a double root in I. At a point of the curve with $I > 0$ we have $\Delta \ne 0$: indeed $F = 0$ gives $(r - \mathcal{R}_0\sigma)\Delta = pI + q > 0$. So the critical points with $I > 0$ are exactly the folds.

6.5 Convexity

Theorem 6.4 (Convexity of $r(I)$)**.** *If $B(1) < KD_1$ (equivalently $\beta > \beta^*$) then $r''(I) > 0$ on $[I_a, I_b]$. If $B(1) = KD_1$, the same holds on $(I_a, I_b]$.*

Proof. Let $n = pI + q$ and $h = \Delta/n = 1/(r - \mathcal{R}_0\sigma)$. By Lemma 6.1, $n > 0$ and $\Delta > 0$ on $[I_a, I_b]$, so $h > 0$ there. Polynomial division gives

$$h = -\frac{A}{p}\, I + e + \frac{f}{pI + q}, \qquad e = \frac{Aq - up}{p^2}, \quad f = M - qe = \frac{Mp^2 - Aq^2 + upq}{p^2}.$$

Substituting $q = MD_1$ and using (6) ($AM = pK$, $p + uD_1 = L_1$),

$$Mp^2 - Aq^2 + upq = Mp\big(p - KD_1^2 + uD_1\big) = Mp\big(L_1 - KD_1^2\big) = MpD_1\big(B(1) - KD_1\big),$$

so

$$f = \frac{MD_1}{p}\,\big(B(1) - KD_1\big), \qquad h'' = \frac{2fp^2}{(pI + q)^3}. \tag{20}$$

Since $r - \mathcal{R}_0\sigma = 1/h$,

$$r'' = \frac{2h'^2 - h\,h''}{h^3}.$$

If $B(1) < KD_1$ then $f < 0$, so $h'' < 0$ and the numerator is > 0. If $B(1) = KD_1$ then $f = 0$, $h' = -A/p \neq 0$, and the numerator is $2h'^2 > 0$. □

6.6 Shape of the branch and type of bifurcation

Theorem 6.5 (Shape and bifurcation type)**.** *Let $B(1) < KD_1$. Then $r(I)$ is strictly convex on $[I_a, I_b]$ with $r(I_a) = r(I_b) = \mathcal{R}_0$ and a unique minimizer I_+ (Proposition 6.2). Moreover:*

- *(a) If $B(1) < 0$ then $I_+ > 0$ and $r_{\min} = r(I_+) = r_2$, the larger root of N. The bifurcation at $r = 1$ is backward: $r(0) = 1$ and $r(I_+) = r_2 < 1$.*
- *(b) If $0 < B(1) < KD_1$ then $I_+ < 0$, the branch is increasing on $I \geq 0$, and the bifurcation is direct; the fold is hidden at a negative I. For $B(1) = 0$ the minimum is at $I = 0$ with $r_{\min} = 1$ (degenerate transcritical point).*
- *(c) For every $r \in (r_{\min}, \mathcal{R}_0)$ the equation $r(I) = r$ has exactly two solutions in $[I_a, I_b]$.*

In all cases $r(I_+)$ is a root of N (a double root of the quadratic in I forces $B^2 = 4AC$).

Proof. Strict convexity, the boundary values and Proposition 6.2 (the other critical point I_- lies outside) give a unique minimizer I_+ in (I_a, I_b), and the sign statements follow from (19). Item (c) follows from strict convexity.

(a) Let (r_2, I_{f}) be the fold of Theorem 5.2, with $I_{\mathrm{f}} = -B(r_2)/(2A) > 0$. It is a point of the curve with $F_I = 0$ and $I_{\mathrm{f}} > 0$, hence a critical point of $r(I)$ by Remark 6.3. The only critical points are $I_\pm$ with $I_- < 0$, so $I_{\mathrm{f}} = I_+$ and $r(I_+) = r_2$. Then r decreases from $r(0) = 1$ as I leaves 0.

(b) $I_+ \leq 0$ implies r is increasing on $[0, I_b]$. □

Remark 6.6 (The case $\beta < \beta^*$). If $B(1) > KD_1$, $r(I)$ has a pole inside $[I_a, I_b]$ (Lemma 6.1(iii)), there is no critical point, and r is monotone on the physical piece. Convexity is not guaranteed there (Section 7).

7 The positive branch for $r \in [1, \mathcal{R}_0]$ in the direct case

Throughout this section $B(1) \geq 0$, i.e. $\beta \leq \beta_0$, so that the positive branch emanates from $I = 0$.

7.1 Existence and monotonicity

For $r \in (1, \mathcal{R}_0]$ we have $C < 0$ by (9), so the quadratic has exactly one positive root

$$I^+(r) = \frac{-B(r) + \sqrt{B(r)^2 - 4AC(r)}}{2A}.$$

Since $B(1) \geq 0$ and $C(1) = 0$, $I^+(1) = 0$; also $I^+(\mathcal{R}_0) = I_b$.

Proposition 7.1 (Monotonicity)**.** *I^+ is continuous on $[1, \mathcal{R}_0]$ and strictly increasing. For $B(1) > 0$,*

$$I^{+\prime}(1) = \frac{-C'(1)}{B(1)} = \frac{M}{D_1 B(1)} > 0, \tag{21}$$

while for $B(1) = 0$ the tangent at $r = 1$ is vertical. (On $(1, \mathcal{R}_0]$ the statement holds for every $\beta > 0$.)

Proof. On the curve $F(r, I^+(r)) = 0$ we have $dI^+/dr = -F_r/F_I$. Here $F_r = B'(r)I + C'(r) < 0$ for $I > 0$, by (7) and (8), and $F_I = 2AI^+ + B = \sqrt{B^2 - 4AC} > 0$ at the larger root. So $dI^+/dr > 0$. At $r = 1$, $C'(1) = -M/D_1$ and $F_I = B(1)$, which gives (21); if $B(1) = 0$ then $F_I = 0$ at $(1, 0)$ and the slope is infinite. □

Consequently the inverse function $r(I)$ of (15), restricted to $[0, I_b]$, is strictly increasing with $r(0) = 1$, $r(I_b) = \mathcal{R}_0$, and

$$I^{+\prime\prime}(r) = -\frac{r''(I)}{r'(I)^3}, \qquad r'(I) > 0, \tag{22}$$

so I^+ is concave iff $r(I)$ is convex on $[0, I_b]$.

7.2 Concavity for $\beta \geq \beta^*$

Theorem 7.2 (Concavity)**.** *If $0 < B(1) \leq KD_1$ (i.e. $\beta^* \leq \beta < \beta_0$), then I^+ is strictly concave on $[1, \mathcal{R}_0]$. The same holds for $B(1) = 0$ ($\beta = \beta_0$), where $I^+(r) \sim c\sqrt{r-1}$ near $r = 1$.*

Proof. By Theorem 6.4, $r''(I) > 0$ on $(I_a, I_b] \supset [0, I_b]$; apply (22). For $B(1) = 0$ we have $I_+ = 0$ and $r'(0) = 0$, $r''(0) > 0$, which gives the square-root behavior. □

7.3 The case $\beta < \beta^*$: concave, or convex then concave

Assume $B(1) > KD_1$. By Lemma 6.1, a pole may lie in $[I_a, 0)$, but on $[0, I_b]$ we have $\Delta > 0$ (it is concave with $\Delta(0) = M > 0$ and $\Delta(I_b) > 0$) and $n = pI + q > 0$, so $r(I)$ is smooth on $[0, I_b]$.

Sign of r''. Let $t = pI + q > 0$ and $h = \Delta/n = 1/(r - \mathcal{R}_0\sigma) > 0$. From the proof of Theorem 6.4, with $I = (t - q)/p$,

$$h = -\alpha t + c_0 + \frac{f}{t}, \qquad \alpha = \frac{A}{p^2}, \quad c_0 = \frac{2Aq - up}{p^2}, \quad f = \frac{MD_1}{p}\big(B(1) - KD_1\big) > 0.$$

Then $h' = -p(\alpha + f/t^2)$, $h'' = 2p^2 f/t^3$ and

$$2h'^2 - h\,h'' = 2p^2\Big[\alpha^2 + \frac{3\alpha f}{t^2} - \frac{c_0 f}{t^3}\Big] = \frac{2p^2}{t^3}\,\psi(t), \qquad \psi(t) := \alpha^2 t^3 + 3\alpha f\, t - c_0 f.$$

Since $r'' = (2h'^2 - hh'')/h^3$ and $h, t > 0$,

$$\operatorname{sign} r''(I) = \operatorname{sign} \psi(pI + q). \tag{23}$$

Because $\alpha, f > 0$, $\psi'(t) = 3\alpha^2 t^2 + 3\alpha f > 0$: ψ is strictly increasing. Hence r'' changes sign at most once on $[0, I_b]$, and only from negative to positive as I increases.

The criterion at $r = 1$. Differentiate $F(r, I^+(r)) = 0$ twice at $(r, I) = (1, 0)$:

$$F_{rr} + 2F_{rI}\, I^{+\prime} + F_{II}\,(I^{+\prime})^2 + F_I\, I^{+\prime\prime} = 0,$$

with $F_{rr} = C''(1) = 2M/D_1^2$ (the term $B''I$ vanishes at $I = 0$), $F_{rI} = B'(1) = -p/D_1^2$, $F_{II} = 2A$, $F_I = B(1)$. Put $J = I^{+\prime}(1) = M/(D_1 B(1))$. Using $AM = pK$,

$$C'' + 2B'J + 2AJ^2 = \frac{2}{D_1^2}\Big[M - pJ + AD_1^2 J^2\Big] = \frac{2M}{D_1^3 B(1)^2}\,\Theta,$$

and therefore

$$I^{+\prime\prime}(1) = -\frac{2M\,\Theta}{D_1^3 B(1)^3}, \qquad \Theta := D_1 B(1)^2 - p\,B(1) + pKD_1. \tag{24}$$

By (22), $\operatorname{sign} r''(0) = \operatorname{sign}\Theta$, and since $h(0) = M/q > 0$, (23) gives $\operatorname{sign}\psi(q) = \operatorname{sign}\Theta$.

Theorem 7.3 (Concavity versus one inflection)**.** *Let $B(1) > KD_1$ (i.e. $0 < \beta < \beta^*$).*

(i) If $\Theta \geq 0$, then I^+ is strictly concave on $(1, \mathcal{R}_0]$.

(ii) If $\Theta < 0$, then I^+ is strictly convex near $r = 1$. It has at most one inflection point $r_* \in (1, \mathcal{R}_0)$: I^+ is convex on $[1, r_*]$ and concave on $[r_*, \mathcal{R}_0]$; if no inflection occurs in $(1, \mathcal{R}_0)$, I^+ is convex on all of $[1, \mathcal{R}_0]$.

In particular $\Theta < 0$ is possible only if $p > 4KD_1^2$ and $B(1)$ lies strictly between the roots

$$B_\pm = \frac{p \pm \sqrt{p^2 - 4pKD_1^2}}{2D_1}, \qquad \text{with } B_- > KD_1.$$

Proof. (i) If $\Theta \geq 0$ then $\psi(q) \geq 0$, and ψ is strictly increasing, so $\psi > 0$ for $t > q$, i.e. $r'' > 0$ on $(0, I_b]$; apply (22). (ii) If $\Theta < 0$ then $\psi(q) < 0$, so $r'' < 0$ near $I = 0$ and I^+ is convex there. By monotonicity of ψ there is at most one zero $t_0 = pI_0 + q$; if $I_0 \in (0, I_b)$ then $r'' < 0$ on $[0, I_0)$ and $r'' > 0$ on $(I_0, I_b]$, which gives the stated pattern with $r_* = r(I_0)$.

For the last claim, Θ is a quadratic in $B(1)$ with $\Theta(0) = pKD_1 > 0$ and $\Theta(KD_1) = K^2 D_1^3 > 0$. $\Theta < 0$ somewhere requires real roots $B_\pm$, i.e. $p > 4KD_1^2$, and then $\Theta < 0$ exactly on (B_-, B_+). Since KD_1 lies outside this interval, either $KD_1 \geq B_+$ or $KD_1 \leq B_-$; the first is impossible because $B_+ \geq p/(2D_1) > 2KD_1$. Hence $KD_1 < B_-$ (and the case $\Theta < 0$ with $B(1) \leq KD_1$ never occurs, in agreement with Theorem 7.2). □

8 Numerical verification

All statements above are proved analytically; random-parameter experiments (positive parameters, $0 < \epsilon < 1$, $\mathcal{R}_0\sigma < 1 < \mathcal{R}_0$) were used to check them and to catch algebraic errors.

Test	Samples	Outcome
Roots of N; parameters in $(0.01, 3)$	200,000	981 with real roots, all in $(\mathcal{R}_0\sigma, 1)$; none negative or of mixed sign
Roots of N; log-uniform in $(10^{-3}, 10^2)$	2,000,000	284,909 with real roots; 4 apparent exceptions at $r \approx 1$, attributed to floating-point error at $B(1) \approx 0$
Convexity of $r(I)$ on $[I_a, I_b]$	200,000	all 26,245 cases with $B(1) < KD_1$ convex with no pole; all 173,755 cases with $B(1) > KD_1$ had a pole in $[I_a, I_b]$
Positive branch, $B(1) \geq 0$	53,634	I^+ always increasing; always concave when $B(1) \leq KD_1$; $\Theta < 0$ in 3,705 cases ($\approx 7\%$)

In the last test, 23 samples disagreed with the sign of Θ; the six examined had $B(1)/(KD_1) > 2000$ and second differences at round-off level, so these are attributed to finite-difference noise (the rest were not inspected). Formula (24) was checked against finite differences for $\beta = 3$ (predicted 2.570, finite difference 2.575) and $\beta = 0.3$ (both -1.5588), and (19) against the direct computation of the critical point for $\beta = 8$ and $\beta = 5.95$ ($I_+ = 0.02036$ and -0.00213 respectively).

9 Concluding remarks

1. The threshold β^* tells when N has real roots; the backward bifurcation itself is governed by β_0 (the sign of $B(1)$). For $\beta^* \le \beta \le \beta_0$ the fold exists but sits at $I \le 0$ and is not physical.

2. Under $\mathcal{R}_0\sigma < 1 < \mathcal{R}_0$ the backward bifurcation is *always* folded: the positive branches exist exactly on $(r_2, 1]$, and a unique crossing or a persistent pair down to $r = \mathcal{R}_0\sigma$ never occurs.

3. The value I_+ of the minimizer of $r(I)$ equals $-B(1)/A$ times a factor in $(0, \frac{1}{2})$ when $B(1) < 0$, so it is not simply proportional to $B(1)$, but its sign is $-\operatorname{sign} B(1)$ exactly.

4. The positive branch of the direct case is always increasing; its concavity is guaranteed for $\beta \ge \beta^*$, while for $\beta < \beta^*$ at most one inflection point can occur and the sign of Θ decides whether the branch starts convex.